

Exercise 6: Potentials 1 and Electric Fields in Matter 1

Question 1:

(a) Fill in how the potential $V(r)$ and the electric field magnitude $E(r)$ scale with r for each multipole:

Name	Leading far-field potential $V(r)$	Leading far-field field $E(r)$
Monopole	$V \propto \underline{\hspace{2cm}}$	$E \propto \underline{\hspace{2cm}}$
Dipole	$V \propto \underline{\hspace{2cm}}$	$E \propto \underline{\hspace{2cm}}$
Quadrupole	$V \propto \underline{\hspace{2cm}}$	$E \propto \underline{\hspace{2cm}}$
Octupole	$V \propto \underline{\hspace{2cm}}$	$E \propto \underline{\hspace{2cm}}$

(b) Give the condition on r under which the pure dipole formulas for V and $\mathbf{E}$ are valid approximations to the physical dipole.

(c) Explain (briefly, conceptually) why the field-line picture of a physical dipole differs near the charges even if it matches far away.

Exercise 2:

Considering the theoretical investigations of Enrico Clementi (table below) that gives the atomic radii (in picometers) of elements. Pick 5 of them and calculate their atomic polarizability (α). For your calculations consider the ‘primitive model’ of the book Example 4.1. Calculate the polarizability of Li and Na. Compare your findings to the values of Table 4.1 in the book?

Group (column)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period (row)																		
1	H 1																	He 2
2	Li 3	Be 4											B 5	C 6	N 7	O 8	F 9	Ne 10
3	Na 11	Mg 12											Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
4	K 19	Ca 20	Sc 21	Ti 22	V 23	Cr 24	Mn 25	Fe 26	Co 27	Ni 28	Cu 29	Zn 30	Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36
5	Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42	Tc 43	Ru 44	Rh 45	Pd 46	Ag 47	Cd 48	In 49	Sn 50	Sb 51	Te 52	I 53	Xe 54
6	Cs 55	Ba 56	* Lu 71	Hf 72	Ta 73	W 74	Re 75	Os 76	Ir 77	Pt 78	Au 79	Hg 80	Tl 81	Pb 82	Bi 83	Po 84	At 85	Rn 86
7	Fr 87	Ra 88	** Lr 103	Rf 104	Db 105	Sg 106	Bh 107	Hs 108	Mt 109	Ds 110	Rg 111	Cn 112	Nh 113	Fl 114	Mc 115	Lv 116	Ts 117	Og 118
			* La 57	Ce 58	Pr 59	Nd 60	Pm 61	Sm 62	Eu 63	Gd 64	Tb 65	Dy 66	Ho 67	Er 68	Tm 69	Yb 70		
			** Ac 89	Th 90	Pa 91	U 92	Np 93	Pu 94	Am 95	Cm 96	Bk 97	Cf 98	Es 99	Fm 100	Md 101	No 102		

Hint: What assumptions will you consider? What is the volume of an atom in this case and why? Bring your results to the units indicated by table 4.1 in the book.

Question 3:

A dielectric can become polarized in two main ways:

1. induced dipoles in atoms/nonpolar molecules ($\mathbf{p} = \alpha \mathbf{E}$), and
2. alignment of permanent dipoles in polar molecules (torque $\mathbf{N} = \mathbf{p} \times \mathbf{E}$).

(a) Theoretically, if a neutral atom is placed in a weak uniform field, the induced dipole moment points in the same direction as $\mathbf{E}$ and is approximately proportional to $\mathbf{E}$.

- Explain why it makes physical sense that $\mathbf{p}$ increases with $\mathbf{E}$, and why the linear relation is expected only when the field is “not too strong.”
- What does the constant α represent physically, and what kind of atomic “response” is it measuring?

(b) Consider Griffiths’ crude atom model (Example 4.1) for an atom consists of a point nucleus $+q$ at the center of a uniformly charged spherical electron cloud $-q$ of radius a . In an external uniform field $\mathbf{E}$, the nucleus shifts a small distance d relative to the cloud, and equilibrium occurs when the external field balances the internal field from the cloud at the nucleus location. Due to this subtle shift a dipole is induced which can be written as, $\mathbf{p} = (4\pi\epsilon_0 a^3)\mathbf{E}$.

- Without re-deriving it, explain the main physical message of this result: why should polarizability scale like volume ($\sim a^3$)?
- The example remarks that the actual displacement of the electron cloud is extremely small. Conceptually: why is it reasonable in the crude model to assume the electron cloud “retains its spherical shape”?

(c) For asymmetrical molecules like CO_2 the polarizability is represented with polarizability tensor with components α_{ij} . Explain why an asymmetric (or direction-dependent) polarizability can make the induced dipole $\mathbf{p}$ not parallel to $\mathbf{E}$.

(d) A polar molecule has a permanent dipole moment $\mathbf{p} = q\mathbf{d}$. In a uniform electric field the net force on the dipole is zero, but the dipole experiences a torque $\mathbf{N} = \mathbf{p} \times \mathbf{E}$. Explain clearly how you can have zero net force but nonzero torque in a uniform field.

Question 4:

A hydrogen atom (Bohr radius $a_0 \approx 0.5 \text{ \AA} = 0.5 \times 10^{-10} \text{ m}$) is situated between two large parallel metal plates separated by $x = 1 \text{ mm}$. The plates are connected to a 500 V battery.

- a) Assuming the electric field between the plates is uniform, estimate the field magnitude E .
- b) Using the hydrogen polarizability $\frac{\alpha}{4\pi\epsilon_0} = 0.66 \times 10^{-30} \text{ m}^3$, estimate the induced dipole moment $p = \alpha E$.
- c) If the dipole moment is modeled as $p = ed$ with e being the charge of the electron, estimate the displacement d , and find the fraction d/a_0 .
- d) Estimate the voltage required for the induced displacement to become comparable to the atomic size ($d \sim a_0$), meaning the electron is no longer effectively bound.

Question 5:

A very long (effectively infinite) solid dielectric cylinder of radius R has polarization

$$\mathbf{P}(s) = k s \hat{\mathbf{s}} \quad (0 \leq s \leq R),$$

where k is a constant, s is the cylindrical radial coordinate, and $\hat{\mathbf{s}}$ is the outward radial unit vector. There is no free charge anywhere.

- (a) Find the bound volume charge density $\rho_b(s)$.
- (b) Find the bound surface charge density σ_b on the curved surface $s = R$.
- (c) Find the electric field $\mathbf{E}(s)$ inside and outside the cylinder.